Exploring the association of Water Withdrawal and Technology Across Countries

SRN 230713279

Word count 4000

Due Date 8th April 2026

(granted extension)

Table Of Content

1. Introduction 3
2. Research Questions 3
3. The Dataset & Variables Description 3
4. Data Preparation & Cleaning 4
5. Exploratory Data Analysis “EDA” 4
5.1 Variable Distributions 4
5.2 Correlation Analysis 5
5.3 Summary Statistics 5
6. Unsupervised Learning 5
6.1. K-Means Clustering 5
6.1.1 Cluster Profiling 6
7.Supervised Learning: Regression 6
7.1 Data Preparation For Regression: 7
7.1.1 Principal Component Analysis - PCA 7
7.2. The linear Regression Model 7
7.3. Regularization methods: Ridge & Lasso 7
7.4. Random Forest “RF” Regression 8
7.5. Model Comparison - Regression 8
8. Supervised Learning: Classification 9
8.1 Data Preparation For Classification 9
8.2 Logistic Regression 9
8.3 Random Forest Classification 10
8.4 XGBoost Classification 11
8.5 Model Comparison - Classification 11
9. Limitations 12
5. Literature Review 12
10.Recommendations 12
11. References 13
12. Appendix 13

1. Introduction

Nowadays, Technology is growing rapidly around the world. From artificial intelligence (AI), cloud computing, networks to broadband infrastructure & computing powered devices. It is used in every sector in our life. As a result, Governments, firms and households increasingly use them to improve productivity, communication, and the decision-making process.

However, these technologies require large data centers and energy resources, intensively depending on water for cooling systems, increasing the consumption water yearly. As (Yañez-Barnuevo, 2025) mentions, Large data facilities consume as much as 5 million gallons of water daily, which is comparable to the water needs of a small town.

Water is already a scarce resource, if the rapid technologies also contribute to larger water withdrawal, it will add up to the worldwide sustainability challenges and increase the environmental pressures.

This project uses the World Bank World Development Indicators Dataset to study the relationship between technogy adoption and freshwater withdrawal across countries. Using the most recent year with near complete data (2022), the study explores whether countries with higher levels of technological developments tend to have large freshwater withdrawals.

To address this, we apply several machine learning methods. First, clustering is used to explore whether countries naturally group into distinct technology-development profiles and see whether these groups differ in their water withdrawal patterns. Second, regression models are used to quantify the association between technology adoption and freshwater withdrawal while controlling for economic and demographic factors. Finally, classification models are used to assess whether country-level technology and development indicators can distinguish between countries with high and low water withdrawal levels.

2. Research Questions

1. Do countries naturally cluster into technology profiles, and are these profiles associated with different levels of freshwater withdrawal ?

2. Does higher technology adoption significantly associated with higher freshwater withdrawal After controlling for economic, demographic and agricultural factors ?

3. Can we identify countries at risk (high fresh withdrawal) and countries who are safe (low fresh water withdrawal) using classification methods?

3. The Dataset & Variables Description

The World Bank Development Indicators “WDI” dataset contains Over 60,000 Rows (one row per country for all countries and years) and around 1,500 indicators related to economic, social, environmental and technological aspects.

For this analysis, only relevant indicators are selected focusing on the **year 2022**, As it is the most recent year with almost complete data for all countries and variables of interest.

The main variable of interest is **Water Withdrawals**, which measures the total freshwater withdrawn (agricultural, industrial, domestic) as a percentage of a country’s total renewable freshwater resources, serving as the target variable in the analysis.

To study the effect of technology, these proxy variables were used:

- Energy use per capita **("Energy"):** Total primary energy consumption per person.
- Electric power consumption per capita **("Electricity"):** electricity use per person.
- Internet users (% of population) **("Internet"):** Percentage of people using the internet.
- Fixed broadband subscriptions per 100 people **("Broadband"):** Shows infrastructure-based internet access.
- Mobile cellular subscriptions (per 100 people) **("Mobile_Subscriptions"):** measures the number of active mobile phone subscriptions relative to the population size.

Additionally, control variables related to economic and demographic factors are added:

- GDP per capita **("GDP"):** Gross domestic product divided by population.
- Total population **("Population"):** Total number of people in the country.
- Agricultural freshwater withdrawals (% of total) **("Agriculture"):** Share of total freshwater used for agriculture such as irrigation, livestock and aquaculture.

4. Data Preparation & Cleaning

After filtering the dataset to the year **2022**, the data set was converted from tall to wide format, creating one row per country with each indicator in a separate column. The original dataset consisted of 266 countries, but with many missing observations. These rows were entirely removed, producing a dataset with **130 countries** to ensure full variable availability for precise modelling. The variables are then renamed with simple names, as mentioned above, to improve readability and clarity.

5. Exploratory Data Analysis "EDA"

Before building any models, we examined the distributions and relationships among variables to identify required transformations, highly correlated predictors and the unusual values that may affect the analysis in the future models.

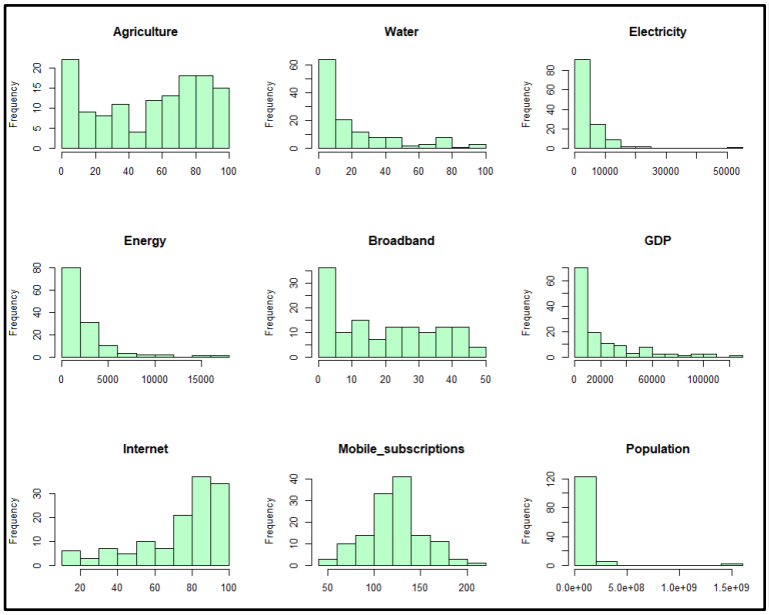


Figure 1: Histograms of the variables

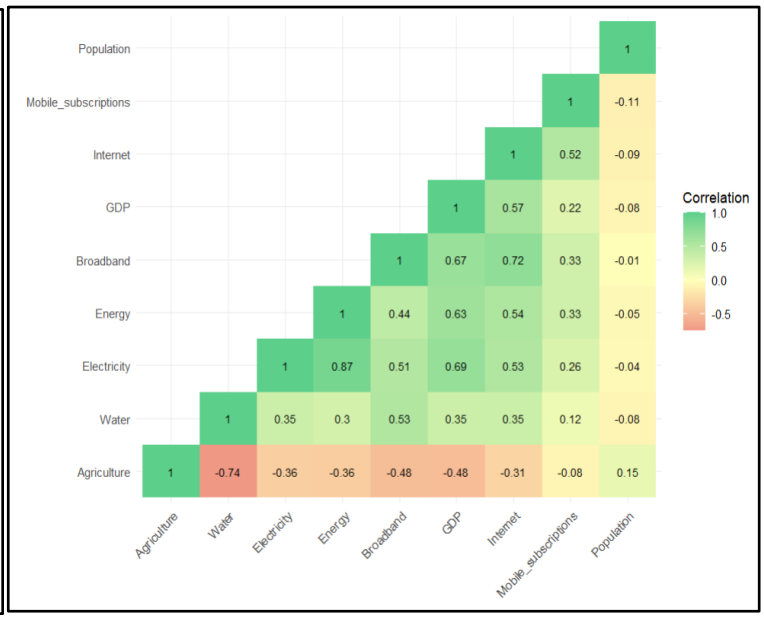


Figure 2 : Correlation matrix of the variables

Figure 1 presents histograms of all continuous variables. Several variables show extreme skewness such as GDP, Water, Energy, Internet and Electricity, particularly, positively skewed. This means that a small number of countries have very high values, while most countries have moderate values.

For example, most countries withdraw little water, while a few withdraw masses, pulling the average water withdrawal up and making the distribution "skew" towards them. On the other hand, Mobile subscriptions and Agriculture are almost symmetric, the countries are evenly spread around the average value.

5.2 Correlation Analysis

To understand how variables move together, we examined correlations among them. A positive correlation means that as one variable increase the other increase, while a negative correlation is the opposite.

The matrix in figure 2 shows that technology indicators are strongly positively correlated, with values ranging between 0.87 and 0.54. Which is expected, countries with high internet access will tend to have high broadband connections, electricity usage, etc.

On the other hand, agriculture water share is negatively correlated with most variables expect population, with the strongest negative relationship observed with Water (-0.74). This **may** indicate that countries with higher agricultural water shares have lower % of water withdrawal and smaller adoption of technology, a reflection of it being a poor/developed economies.

5.3 Summary Statistics

Table 1 displays summary statistic for all variables. The variation across the countries is very high. For example, GPD range from $503 to $123,720, internet users range from 11% to 100% and water withdrawal ranges from zero to 94.2 m^3. This reflects the diversity of each country economically and technologically.

	Variable	Mean	SD	Min	Max
Agriculture	Agriculture	5.182134e+01	3.180713e+01	0.000000e+00	9.813522e+01
Water	Water	2.122997e+01	2.465876e+01	4.676041e-02	9.419721e+01
Electricity	Electricity	4.493504e+03	6.123261e+03	1.446738e+01	5.125876e+04
Energy	Energy	2.378904e+03	2.728113e+03	1.985448e+02	1.668366e+04
Broadband	Broadband	1.925266e+01	1.523520e+01	0.000000e+00	4.825020e+01
GDP	GDP	1.980236e+04	2.556432e+04	5.035145e+02	1.237197e+05
Internet	Internet	7.435094e+01	2.288406e+01	1.112460e+01	1.000000e+02
Mobile_subscriptions	Mobile_subscriptions	1.213325e+02	3.074424e+01	4.581780e+01	2.072780e+02
Population	Population	5.644844e+07	1.799080e+08	3.820030e+05	1.425423e+09

6. Unsupervised Learning

6.1. K-Means Clustering

To identify whether countries naturally group based on technology and how the groups differ in their water withdrawal, we have applied **k-means clustering**.

Using all the continues variables mentioned above – expect country - , we standardized and removed outliers to ensure equal contribution and prevent extreme values from disproportionally effecting the cluster assignment procedure.

To determine the optimal number of clusters "K", we used the "Elbow method". This method produces a graph that can help choosing the **ideal** number of clusters for the dataset.

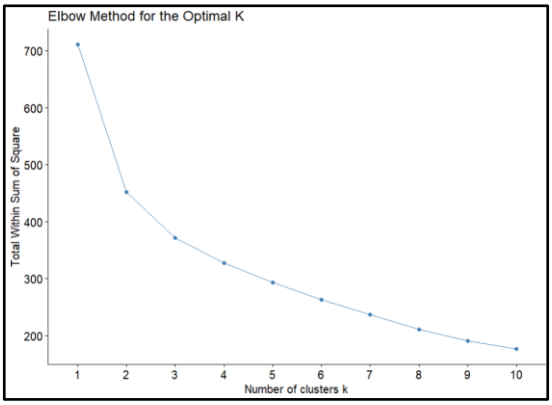


Figure 3: Number of clusters against WSS

It measure how tightly grouped the countries are within their assigned clusters, which is known as the within-cluster sum of squares (WSS). Smaller WSS values indicates tighter, more homogenous clusters and vice versa. As K increases, WSS decreases because more clusters fit better in the data. However, the key is to find the moment at which the **rate** of the decrease starts to slow down, creating an "elbow" so the graph that signify the optimal number of clusters. As shown in figure 3, the elbow occurs at K =, so 3 clusters were created in this analysis.

6.1.1 Cluster Profiling

After focusing on technology variables, Table 2 shows the average values, along with water withdrawal for each cluster and the number of countries in each group.

Cluster	N	Energy	Electricity	Internet	Broadband	Mobile_subscriptions	Water
1	27	579.6	789.3	56.0	4.0	101.8	9.2
2	32	1466.6	2520.9	80.7	18.4	135.6	9.1
3	21	2999.8	5603.6	90.3	33.1	127.1	34.1

The below clusters were identified:

1) Low Development /Tech Adoption Group: contains 27 countries, these are developing economies countries, with limited energy access, weak digital infrastructure and high agriculture dependency. They have the lowest tech adoptions in all technogy variables.

2) Middle Development/Tech Adoption Group: contains 32 countries, the biggest portion of the clusters. These are emerging economies that have substantial energy and digital infrastructure, thus, higher technology adoption than the first group. Moreover, They the most water-efficient group, having the least water withdrawals.

3) High Development/Tech Adoption Group: they are advanced, highly digitalized economies, with 21 countries only, they exhibit the highest technology averages in most indicators, expect Mobile subscriptions, where cluster 2 countries beats.

To look at how water withdrawal differ in each cluster, figure 4 shows the distribution of water withdrawal for each group. First, we notice that cluster 1 and 2 have almost the same water withdrawal levels, even though cluster 2 has noticeably higher technology usage. Suggesting that moderate levels of tech adoption can be achieved without increasing water demand.

On the other hand, cluster 3 shows dramatic increase in water withdrawal, almost four times higher than the other clusters, suggesting the possibility of having nonlinear relationship.

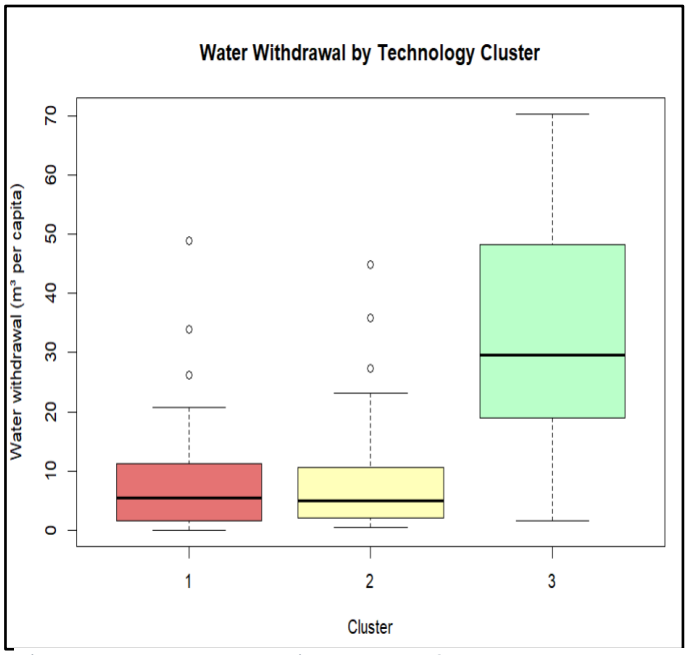


Figure 4: Boxplot of water withdrawal for Clusters

7.Supervised Learning: Regression

While clustering hints for the effect between technology and water withdrawal, it does not quantify the impact of technology directly. This regression task aims to quantify whether higher technology adoptions are associated with significant increase in freshwater withdrawal.

7.1 Data Preparation For Regression:

Before building the regression models, two main transformations were needed to improve the model's performance.

First, as shown earlier in figure 1, water withdrawal, GDP, and population were highly skewed. Therefore, we applied **natural logarithm** transformations, which reduces skewness, improves linearity and limit the impact of outliers. This enables a more uniformed interpretations of proportional differences. Second, Figure 2 shows that technology variables are highly correlated, this prevents the model from isolating individual effects, a problem called "multicollinearity". To solve this, we used a dimensionality reduction technique "Principle Component Analysis" to combine Energy, Electricity, Internet, and Broadband into a single **"technology index".**

7.1.1 Principal Component Analysis - PCA

The analysis aims to reduce the dimensionality of the data by summarizing correlated variables into fewer "components". Each component receives a set of **weights** called "loadings" showing the contributions from original variables and a **percentage** of how much it explains the variations of those variables. Using the weights of the first component, which revealed the highest level of explanation "61", each country received a score of how much technongy it adopts, calculated as follows:

Tech_Index = (Energy_std × 0.4778) + (Electricity_std × 0.4776) + (Internet_std × 0.4878) + (Broadband_std × 0.4435) , Where each variable has been standardized before weighting.

7.2. The linear Regression Model

To examine the relationship between water withdrawal and technology, we estimated a linear regression model predicting log transformed water from technology index, agricultural water share and log transformed population. GDP was excluded as it exhibits large correlation (0.73) with the technology index, creating issues as mentioned above.

The regression showed that countries with higher technology adoption tend to use more water, the coefficient of Tech_index is positive and statistically significant 0.158, $p < 0.009$ respectively. Meaning that holding other variables constant, a one unit increase in technology index is predicted to increase water withdrawal by 17%.

As shown in table 4, the model was evaluated using 10- fold cross validation giving RMSE= 1.037 and cross validated $R^2=0.557$ where the model explains 55.7% of the variations in log_water withdrawals suggesting that the linear model generalizes reasonably.

Moreover, We tested if the model satisfy all the linear regression assumptions and detected mild heteroskedasticity, which means that prediction errors vary with each level of water withdrawal, effecting the statistical significance. However, after applying the heteroskedastic robust standard errors, the significance of the variables haven't change, strengthening the confidence in our findings.

7.3. Regularization methods: Ridge & Lasso

Ridge and lasso are models that extend linear regression by adding a brake "penalty" to the model, where the coefficients shrink towards zero in ridge , and reach to zero in lasso to help weakening/eliminating the unnecessary variables in the regression model, preventing overfitting. Because our sample size is small, we used them as a check of how robust the model is and to confirm the stability of the coefficients with the shrinkage.

As shown in table 3 , Both models produced coefficients and predictive performance very similar to the linear regression model (Ridge RMSE = 1.083, Lasso RMSE = 1.066). There was no severe shrinkage and variable elimination, as lasso retained all the three variables proving that all selected predictors are strong predictors.

Coefficient Comparison Across Regression Models

Predictor	OLS	Ridge	Lasso
Tech_Index	0.158	0.159	0.156
log_population	0.204	0.173	0.200
Agriculture	−0.033	−0.030	−0.033

Table 3

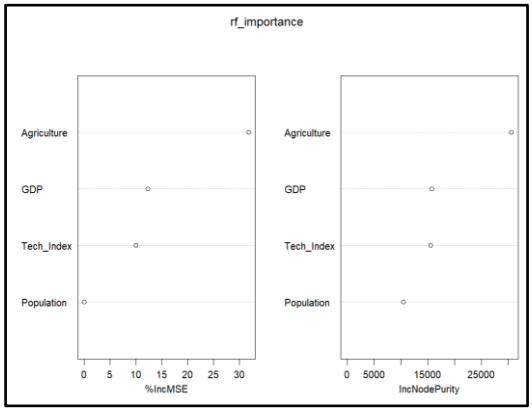


Figure 5

7.4. Random Forest "RF" Regression

The clustering analysis suggested that the relationship might not be linear, due to the fact that water use remained the same from low to medium technology levels , then jumped sharply for the highest tech countries. To detect this, we used Random forests. Unlike linear regression were it assumes a straight line relationship, Random forests creates decision trees and average their predictions allowing complex, nonlinear patterns to be detected.

We trained the model using water withdrawal on its original scale, technology index, GDP, agricultural water share, and population, as it has the ability to identify patterns through correlations & skewness.

As shown in table 4, The model explained 68.3% of the variation in water withdrawal across countries. Outperforming the linear models. Moreover, Agriculture, GDP and Technology index appeared as the strongest predictors (figure 5), supporting the idea that the relationship between water withdrawal and country developments include important nonlinear effects that linear models cannot entirely capture.

7.5. Model Comparison - Regression

Overall, the results from all models were consistent, certainly showing a positive relationship between technology and freshwater withdrawal.

Linear, ridge and lasso produced almost the same coefficients, suggesting that multicollinearity was successfully solved by preforming PCA. Random forests appeared to be the best for explaining the variations As shown in table 4, with highest cross validated R squared .

From prediction perspective, since linear models were fitted on **log transformed water** and random forests fitted on **non-transformed water**, direct comparison of **RMSE** is not valid. Therefore, the predictions of the linear models were back transformed using bias-corrected exponential transformation, the results are reported in table 5.

Indeed, the Random forests predictions appeared to be the best, with the lowest prediction error "RMSE" – a 30.67% error decrease from the linear model-.

From an interpretability perspective, the linear regression model preforms best. It offers clear statistical signicance evidence and a clear coefficients that are easy to communicate to policy makers.

For instance, you can state with 99% confidence that tech_index is significant in predicting water withdrawal and he results are not just a random chance. Even though random forest does indicate the importance of variables, but not the confidence with the results.

Cross-Validated Model Comparison on Water Scale

Bias-corrected back-transformation for log-based models

Model	CV_RMSE	CV_R2
OLS	21.957	0.594
Ridge	17.806	0.625
Lasso	20.113	0.621
Random Forest	15.222	0.683

Table 5

Cross-Validated Model Performance Comparison

10-Fold CV metrics for all predictive models

Model	CV RMSE	CV R^2
OLS	1.037	0.557
Ridge	1.089	0.488
Lasso	1.079	0.497
Random Forest	15.222	0.683

Table 4

As a result, we use linear regression to understand the relationship, Random forest for the most accurate predictions, and ridge/lasso to confirm the robustness.

8. Supervised Learning: Classification

After quantifying how technology is associated with water withdrawal, Policy makers may need a simpler framework to create targeted policies and plan resources efficiently. Hence, this classification task aims to examine whether technology adoptions, economic development and demographic factors can classify countries with **high** freshwater withdrawal and **low** withdrawal levels.

8.1 Data Preparation For Classification

To create the target variable, countries were divided into two groups based on their water withdrawal. Countries with water withdrawal of 25% and above were labelled as “high withdrawal”, while the rest were labelled “low withdrawal”. The classification follows UNEP (2025), where values of water below 25% indicates no water stress, and the values above otherwise. Thus, a binary target was created where high water withdrawal =1 and low water withdrawal =0

For all models, we used the same predictors produced in the regression data preparation section, which are Technology Index, Log GDP, Agricultural Water Share and Log Population, along with the generated binary variable.

8.2 Logistic Regression

The logistic regression is used to predict a target binary variable, producing a probability of each observation to how probable they belong to the target group or not.

Although the model was estimated using multiple of predictors, **figure 6** shows how two important predictors naturally separate between low and high water withdrawal countries. Despite some overlap between the two colors, which justifies the use of extra predictors, This makes us visualize seeking a line or a curve that splits the predictor’s space.

We trained the model using 10-fold cross validation and achieved an AUC of 0.938, which means that the model can correctly assign higher probability for high water withdrawal countries 93.2% of the time.

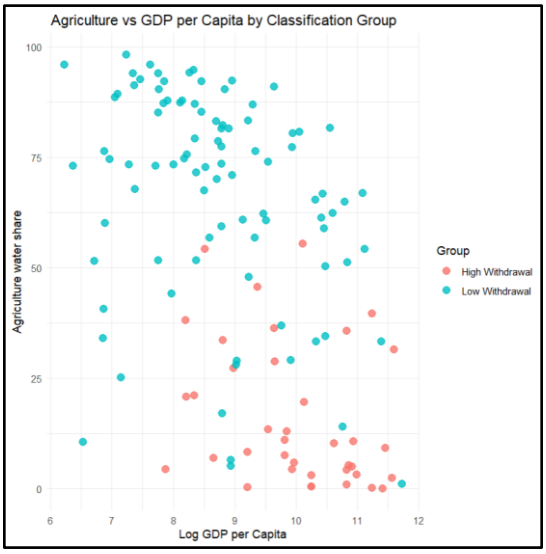


Figure 6

Figure 7 displays the ROC curve, which plots the true positive rate (sensitivity) against the false positive rate (1- specificity) of the model's predictions. The curve is sharp on the top left corner, exceeding the 0.50 baseline of random guessing which indicates high discriminatory power.

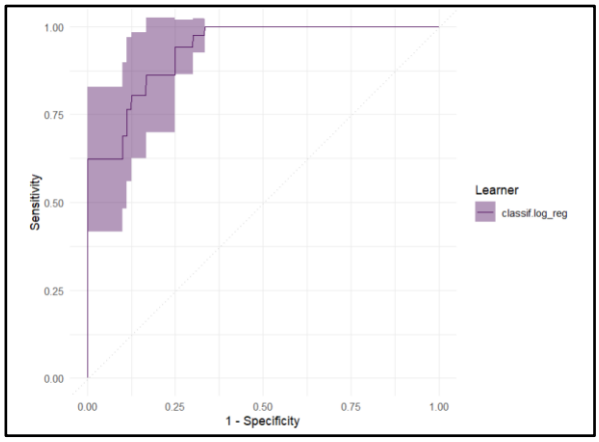


Figure 7

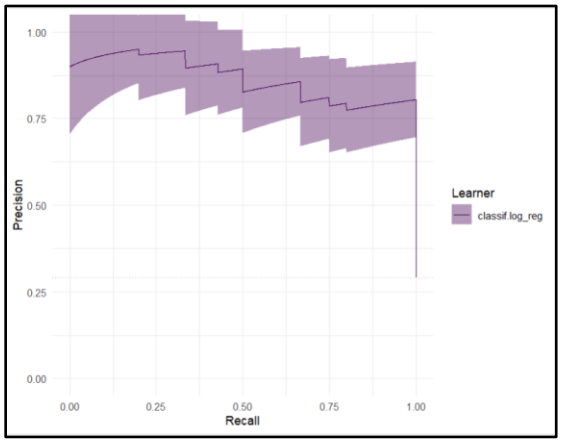


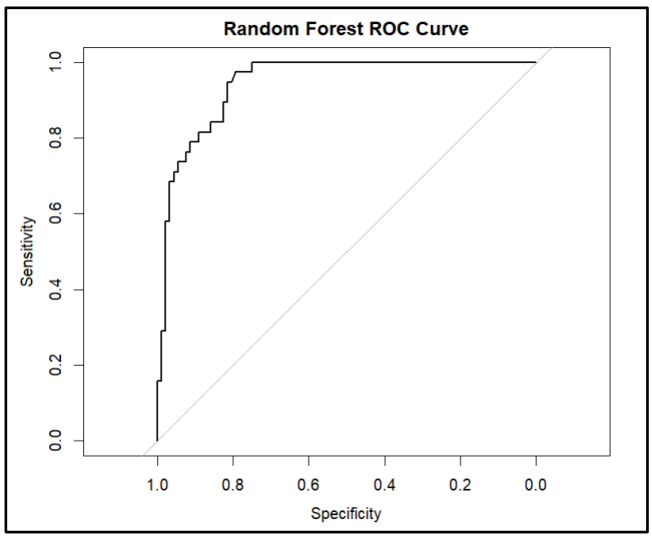
Figure 8

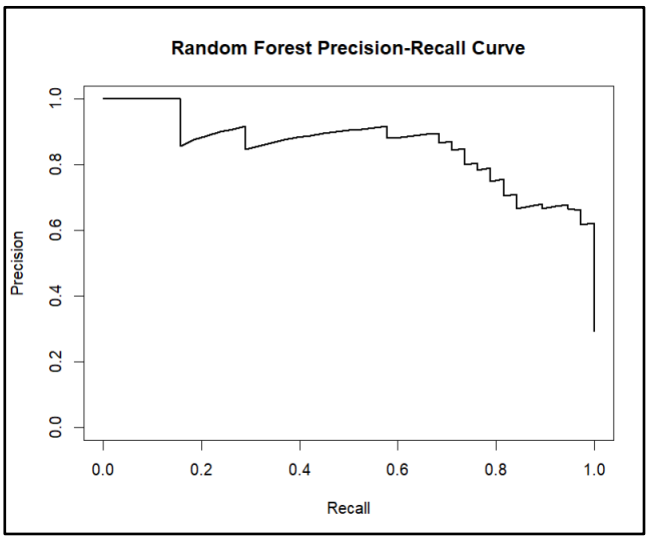
Similarly, the Precision- Recall curve (figure 8) shows that the model have high precision, it reaches 0.85 then declines gradually, where the recall increases instead. This suggests that the model balanced in correctly identifying high withdrawal countries & limiting the false positives.

8.3 Random Forest Classification

Instead of averaging the prediction like in random forest regression, Random forest classification builds decision trees and takes the **majority vote of trees** allowing more flexible and nonlinear classifications.

The model was evaluated using 10-fold cross validation with hyperparameter tuning to discover the values of parameters that gives the best classifications. Both ROC and Precision-Recall curves shows excellent separation, slightly outperforming logistic regression in AUC, indicating that non-linearity exists in the predictors.





8.4 XGBoost Classification

An upgrade of random forest is XGboost, it builds trees sequentially by correcting previous mistakes made by previous trees in creating the new ones, which improves performance especially if the predictors and the target have complex ,nonlinear relationships.

Evaluated using 10-fold cross validation, the model achieved an AUC of 0.958 , slightly higher than random forests (figure 11), Moreover, its Precision-Recall curve maintains its stability from previous models with slightly lower Precision.

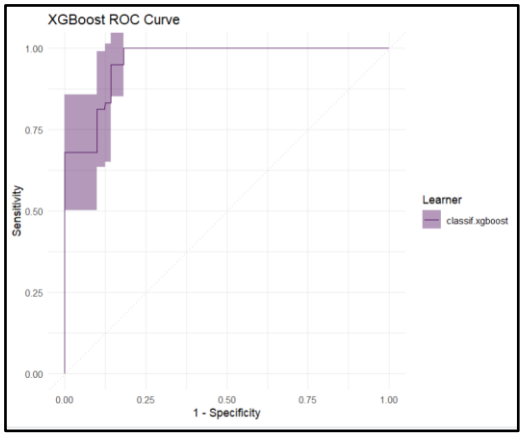


Figure 11

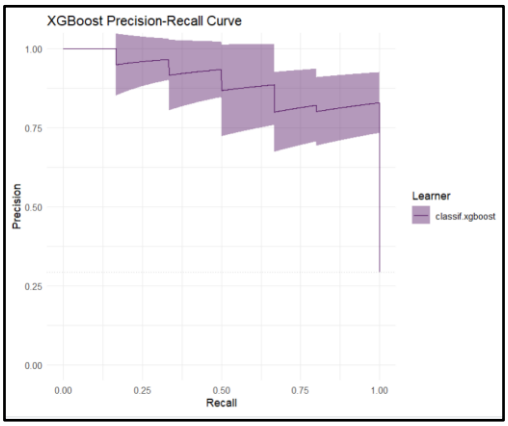


Figure 12

8.5 Model Comparison - Classification

To sum up, table 6 summaries the 4 key indicators for the previously analyzed models. Accuracy was excluded due to its high sensitivity to imbalanced data.

When it comes to the discriminative Ability “ROC-AUC”, XG boost preformed best, correctly classifying the countries 95.8% of the time. Followed by random forest (95.1%), and logistic regression (93.8%). This indicates that XG boost have the best overall ability in distinguishing between high and low water withdrawal.

On the other hand, Precision-Recall showed different patterns. Logistic regression achieved the highest precision 81% meaning, it is correct 81% of the time when it predict high water withdrawal countries. Policy makers who wish to prevent false alarms will find this helpful. In contrast, XGboost has the lowest precision (75.7%), but it compensate by having a high recall (78%) suggesting that it catch more true high withdrawal countries but at a cost of predicting high water withdrawal when a country is not. While the balanced metric “ F1 score” shows that logistic regression outperforms all.

Given these findings, choosing the best model depends on the aim of the policy maker. If the aim is resource efficacy by minimizing false positives, Logistic regression works best. On the other hand, If the aim is to maximize Detection of countries at risk, XGboost works best. Finally, if the aim is to have a balanced model, Random Forests works best. However, in practice according to Occam’s Razor simpler models are preferred when performance differences are small. Making logistic regression the strongest candidate for policy applications.

Classification Model Performance Metrics

Model	ROC	Precision	Recall	F1 Score
Logistic Regression	0.938	0.810	0.790	0.774
Random Forest	0.951	0.784	0.763	0.773
XGBoost	0.958	0.757	0.780	0.746

Table 6

9. Limitations

Although the analysis provides strong evidence on the relationship between technology and water withdrawal, several limitations must be addressed.

First is the availability of data center water consumption datasets, using the proxy variables was helpful, but they don't fully represent the modern technologies, which are likely to be more water intensive. The reason is that there are concerns of transparency, fewer than one third a third of data center operators are monitoring their water wastage, making it difficult to find directly related dataset to measure the impact.

Second is the amount of missing data, even though the dataset has many observations, doing the analysis on the recent "almost completed data" year reduced the sample to 130, which may introduce bias, and creates risk of overfitting the models. Especially in complex models such as Random Forests and XGBoost, as they work best with larger samples.

5. Literature Review

All existing studies agreed that the water footprint of technology in not receiving full attention, partly because it's a recent issue and research is limited due to transparency issues.

Supporting the modelling strategy in this paper, we present a study by Görenekli and Gülbağ (2024) conducting a machine learning analysis on predicting water consumption in Turkey. They used neural networks, support vector machines, random forest and gradient boosting (GBM) along with historical, weather and temporal predictors. Their results show that GBM achieved the highest performance (R^2= 0.881), followed by Random Forest (R^2 = 0.872), proving that ensemble tree-based methods outperform when nonlinear relationships exist in water related data sets. This is consistent with the results of this paper, where the ensemble tree-based methods outperformed them all in regression, and XG boost achieved the highest ROC-AUC value in classification.

Another study is by Li et al. (2023), carrying a principled methodology study to estimate the water footprint of artificial intelligence systems using a framework that combines all influencing factors such as server workload, data center cooling systems, electricity generation and the regional weather conditions to estimate the freshwater demand. The results are consistent with this paper, it shows that water demand of AI centers increase as modelling and computations increase, plus, it predicts the global water withdrawal to achieve a volume of 4.2 to 6.6 billion cubic meters annually by the year 2027, which support the effect identified in this paper.

10.Recommendations

In conclusion, Based on the findings, the following recommendations are proposed for policy maker along with suggesting areas for further research. First, countries that intends to adopt advanced technologies and improve its infrastructure should accompine it with water resource planning. Second, since technology adoptions increase water usage, water efficiency cooling systems, recycled water usage, renewable-powered low-water data centres policies should be employed. Lastly, further research could be done by expanding the study across multiple years to capture the time series impact of water use over time.

11. References

- Yañez-Barnuevo, M. (2025, June 25). *Data centers and water consumption*. Environmental and Energy Study Institute (EESI). https://www.eesi.org/articles/view/data-centers-and-water-consumption

- Mytton, D. (2021). Data centre water consumption. *npj Clean Water*, *4*(1), 11. https://doi.org/10.1038/s41545-021-00101-w

- Ristic, B., Madani, K., & Makuch, Z. (2015). The water footprint of data centers. *Sustainability*, *7*(8), 11260-11284. https://doi.org/10.3390/su70811260

- Li, P., Yang, J., Islam, M. A., & Ren, S. (2025). Making AI less "thirsty": Uncovering and addressing the secret water footprint of AI models. *Communications of the ACM*. Advance online publication. https://doi.org/10.48550/arXiv.2304.03271

- Duignan, B. (2026). Occam's razor. In Encyclopedia Britannica. Retrieved April 8, 2026, from https://www.britannica.com/topic/Occams-razor

- United Nations Environment Programme. (2025). 2.b.1 Level of water stress: Freshwater withdrawal as a proportion of available freshwater resources. *SDG Indicators*. https://sdgs.unep.org/article/2b1-level-water-stress-freshwater-withdrawal-proportion-available-freshwater-resources

12. Appendix

An explanation of some of the machine learning terminologies used is provided below:

- **RMSE**: the root mean square error measures the prediction error, where lower values imply a better model fit.
- **R Squared**: the amount of variation in the dependent variable explained by the predictors.
- **Precision**: the proportion of the predicted positives in the data that the model correctly predicts as positive.
- **Recall (Sensitivity)** : the proportion of the real positive occurrences in the data that the model correctly predict as positive.
- **F1 score**: mean measure that combine both precision and recall.
- **Specificity**: the proportion of the real negative occurrences in the data that the model correctly predicts as negative.
- **AUC**: it is the probability that the model will mark a randomly selected positive instance higher than a randomly selected negative instance.
- **Heteroskedastic-robust standard errors**: it's a type of standard errors used when the variance of residuals is not constant around predicted values.
- **Overfitting**: it happens when the model cannot predict new data that has not been trained on previously as a result of memorizing the training data,
- **Bias-corrected exponential transformation**: A method for back-transforming logged predictions to original units while adjusting for systematic underestimation introduced by the log transformation.